ModelIt!

Student Activity Pack

Why Earthquakes Hit Some Cities Harder

Modeling Seismic Energy, Ground Conditions, and Urban Vulnerability

Grade 09L1 | NGSS HS-ESS2-1 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Why do you think some buildings collapse in an earthquake while others right next to them are fine?**

2. **What causes earthquakes, and why do they happen in some places but not others?**

3. **If you were designing a building in an earthquake zone, what features would you include to keep it standing?**

4. **Do you think the ground beneath a building matters during an earthquake? Why or why not?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Tectonic Stress | Fault Slip | Seismic Wave Energy | Building Vulnerability | Ground Amplification

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Bedrock City

Moderate earthquake, city built on solid bedrock with modern codes

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Soft Soil City

Major earthquake, city built on soft sediment with older construction

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Two Cities Comparison

Same earthquake affecting two cities with different ground and building conditions

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Aftershock Frequency — The number and intensity of smaller earthquakes that follow a major event, which can collapse already-weakened structures and hamper rescue efforts for days to weeks after the mainshock

Tsunami Potential — The risk of ocean displacement when submarine earthquakes cause sudden vertical seafloor movement, generating tsunami waves that can travel across entire ocean basins at jet speed

Infrastructure Interdependency — The cascading failure of interconnected systems — a ruptured gas line causes fires, a broken water main prevents firefighting, downed power lines block roads — multiplying damage beyond direct seismic effects

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do two cities the same distance from an earthquake experience completely different levels of destruction? Use evidence from your simulation to support your answers.

1. **Why is ground amplification often a bigger factor in earthquake destruction than the earthquake's magnitude?**

2. **What does your model reveal about why wealth and poverty determine earthquake survival as much as geology?**

3. **How does the timescale difference between stress buildup (centuries) and stress release (seconds) affect earthquake preparedness?**

4. **Why might a city that hasn't experienced an earthquake in a long time actually be at HIGHER risk?**

5. **What are the limitations of modeling earthquake damage with only five components?**

8 STEM Challenge: Design an Earthquake Risk Assessment System

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YOUR ENGINEERING MISSION

Design a risk assessment and early warning system that identifies the most vulnerable areas in a city and provides actionable information to emergency responders and residents before, during, and after an earthquake.

THE SCENARIO

Your city sits near an active fault zone that hasn't produced a major earthquake in 120 years — meaning enormous tectonic stress has accumulated. The city council has hired your team to create a seismic risk assessment that maps vulnerability across different neighborhoods based on soil type, building age, and proximity to the fault. Your system must identify which areas face the highest risk and recommend prioritized actions.

Engineering Design Thinking

1. WHAT DATA SOURCES WOULD YOU USE TO MAP SOIL CONDITIONS, FAULT PROXIMITY, AND BUILDING TYPES ACROSS THE CITY?

2. HOW WOULD YOU CREATE A RISK SCORING SYSTEM THAT COMBINES GROUND AMPLIFICATION POTENTIAL WITH BUILDING VULNERABILITY?

3. WHAT WOULD YOUR EARLY WARNING SYSTEM COMMUNICATE, AND HOW MANY SECONDS OF WARNING WOULD DIFFERENT ZONES RECEIVE?

4. HOW WOULD YOU PRIORITIZE WHICH BUILDINGS OR NEIGHBORHOODS NEED RETROFITTING FIRST?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Seismologists study earthquakes and seismic waves using networks of sensors around the world, earning $60,000–$130,000/year. Structural Engineers design buildings to withstand earthquake forces using specialized materials and damping systems, earning $70,000–$140,000/year. Emergency Management Directors coordinate disaster preparedness and response, earning $75,000–$150,000/year.